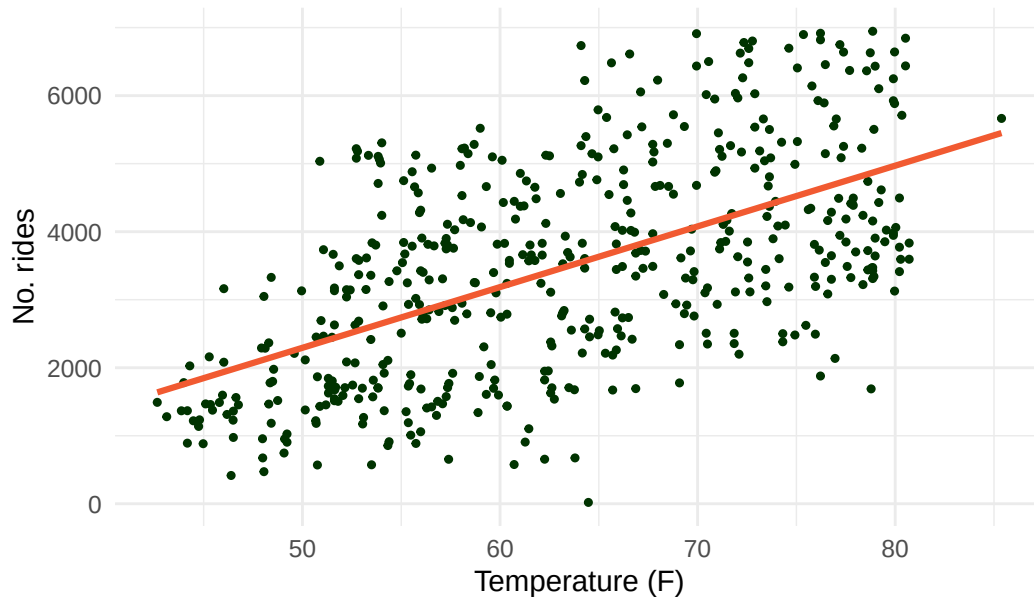


Day 03: Inference for SLR

Name: _____

Data for today

Capital Bikeshare is a bike sharing service in the Washington, D.C. area. We're analyzing the relationship between number of rides taken and temperature.



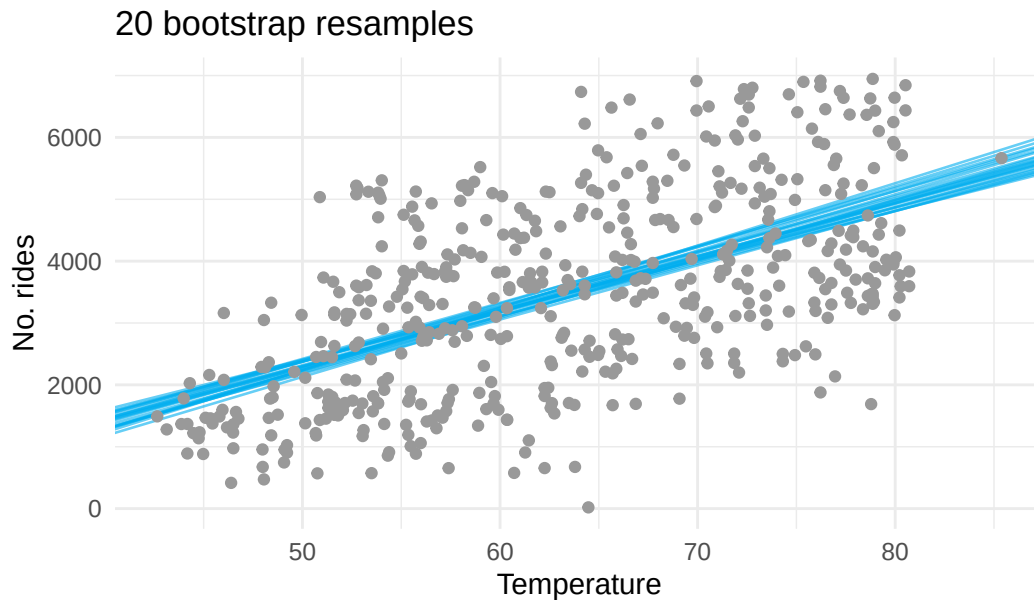
Last time

1. $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ where $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$
2. Scatterplots + correlation
3. Fit the regression line in R and look at a `summary()`
4. Making confidence intervals using `confint`

Today: moving from *descriptive* to _____

Sampling variability

Every new sample of riders gives us a slightly different fitted line. That variability in our _____ (not the data) is what lets us build confidence intervals and hypothesis tests.



Sampling distribution of a coefficient

We use the **t distribution** to model the sampling distribution of a regression coefficient.

- Degrees of freedom: $df = \text{_____}$
- Mean: $= \hat{\beta}_i$
- Standard deviation: $SE(\hat{\beta}_i) = \frac{\hat{\sigma}}{\sqrt{\sum (x_i - \bar{x})^2}}$

Confidence interval (general form)

statistic $\pm$ critical value $\times$ standard error

$$\hat{\beta}_i \pm t^* \cdot SE(\hat{\beta}_i)$$

i Critical value

t^* is the value that cuts off the right tail area that matches your confidence level – e.g. for a 95% CI, find t^* so that the two-sided tail area is 0.025 on each side, using `qt()`.

Hypothesis test (general form)

$$H_0 : \beta_i = \text{null value} \quad \text{vs.} \quad H_a : \beta_i \begin{cases} \neq \\ < \\ > \end{cases} \text{null value}$$

$$\text{test statistic} = \frac{\text{estimate} - \text{null value}}{SE} = \frac{\hat{\beta}_i - \text{null value}}{SE(\hat{\beta}_i)}$$

Then find the appropriate tail area(s) under the t-distribution with $df = n - 2$.

Coefficient table

R's `summary()`/`tidy()` output gives us a coefficient table:

term	estimate	std. error	t	p-value
Intercept	-2168.774	363.642	-5.96	<0.001
temp_actual	89.234	5.670	15.74	<0.001

💡 Try it

Verify the test statistic for $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$ using the table above.

Example: 90% confidence interval for β_1

$$\hat{\beta}_i \pm t^* \cdot SE(\hat{\beta}_i)$$

$t^* = 1.648$ (from `qt(0.95, df = nrow(bikes) - 2)`)

$$89.23 \pm 1.648 \cdot SE(\hat{\beta}_1) \Rightarrow (79.89, 98.58)$$

Interpretation: We are 90% confident that for every 1°F increase in temperature, the number of rides changes by between 79.9 and 98.6, on average.

💡 General template (reusable for any CI)

We are _____% confident that for every 1 _____ increase in _____, _____ by between _____ and _____.

Your turn: on paper

i Note

Work the hypothesis test and confidence interval for this example by hand before checking your answers against the worked example above.

Next: intro to R and RStudio
